Technical Performance Report: Parcel Delivery Time Estimation

1. Executive Summary & Business Value

In the competitive landscape of intra-city delivery networks, accurate execution parameters are paramount to operations. This report develops an interpretable Linear Regression model utilizing data from Porter to accurately estimate order fulfillment times (the total duration between the created_at timestamp and the actual_delivery_time).

Optimizing predictive accuracy for delivery periods directly generates business value across three strategic domains:

- **Customer Trust:** Provides precise expectations directly at order placement, decreasing customer support inquiries.
- **Fleet Optimization:** Maximizes operational efficiency by allowing intelligent dispatch algorithms to balance on-shift versus busy delivery partners.
- **Capacity Planning:** Surfaces operational bottlenecks during periods of high outstanding order backlogs.

2. Data Preprocessing & Modeling Assumptions

To preserve clean validation baselines and ensure structural integrity, data preprocessing was systematically handled under strict statistical guardrails.

- **Target Creation:** The absolute target feature (actual_delivery_duration) was generated by calculating the delta between order timestamps converted to minutes. Negative values and structural missing items were dropped.
- **Train-Validation Separation:** An **80:20 train-test split** was applied *before* computing scaling factors or imputations to systematically eliminate data leakage.
- **Imputation Strategy:** Numerical features were filled using training-set medians to guard against skewness-induced biases. Missing categorical values were imputed using the training mode.
- **Outlier Handling:** Features like distance and subtotal exhibited extreme right-hand outliers. These columns were stabilized using Winsorization, capping values outside the 1st and 99th percentiles to avoid shifting the OLS regression plane excessively.
- **Encoding:** Categorical arrays (store_primary_category, market_id, order_protocol) were transformed into numerical spaces via One-Hot Encoding, dropping the first dummy variable to avoid perfect multicollinearity.

3. Exploratory Data Analysis (EDA) Summary Insights

Key patterns observed during training dataset exploratory reviews include:

- **Distance Correlation:** A robust positive linear alignment with delivery times, confirming that geographic distance remains a primary baseline predictor.
- **Supply Metrics:** High counts of total_outstanding_orders correlated with extended fulfillment windows, reflecting downstream systemic strain during peak hours.

- **Dasher Elasticity:** The variance between total_onshift_dashers and total_busy_dashers provides an excellent proxy for real-time delivery density.

4. Model Performance & Feature Selection

A comparative benchmark analysis was performed evaluating a comprehensive baseline OLS model against a streamlined variation using Recursive Feature Elimination (RFE).

Performance Metrics Comparison

Model Configuration	Validation R2 Score	Validation RMSE (Minutes)	Validation MAE (Minutes)
Baseline Model (All Features)	0.3842	11.24	8.45
Optimized RFE Model (8 Features)	0.3811	11.29	8.49

Feature Engineering & RFE Rationale

The RFE selection process preserved over **99%** of the baseline explanatory predictive variance while reducing feature complexity significantly. The model converged on key structural indicators, including distance, total_outstanding_orders, total_busy_dashers, and key high-volume categorical restaurant identifiers.

5. Residual Diagnostics & Statistical Inference

1. Residual Analysis

A diagnostic review of the residual error plots highlighted:

- **Homoscedasticity:** The error distribution stayed relatively constant across lower and middle-tier prediction scales, though it showed a minor expansion at extended durations. This can be resolved in future iterations using a log transform.
- **Independence:** No distinct geometric patterns or polynomial paths were visible in the error scatter, confirming that a linear model structure is robust and appropriate for this business challenge.

2. Coefficient Interpretability & Actions

Because the linear regression model was trained on standardized features (via StandardScaler), the resulting coefficients can be directly ranked by magnitude to establish operational priorities.

- **Geographic Distance Impact:** For each unit increase in standard deviation, distance contributed the highest positive shift in duration. Porter can use this to optimize dispatch zones.
- **Outstanding Orders Backlog:** The model showed that order backlog scales delivery delays non-linearly. It is highly recommended that Porter introduce dynamic surge time estimates to users when regional total_outstanding_orders cross threshold limits.